Universal gravitation

antocciphysics | TEACHER KEY | CPA

Learning target: Predict how gravitational force changes with mass and separation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 1; 3 course days later.*
Changing heading or a curved sequence of positions shows direction changes; velocity includes direction.

02. Explain a worked example (worked)
Center separation r then 2 r; each body's force drops from 20 to 5 N, with equal opposing forces within each pair.

03. Look and explain (model)
Distance is between centers; equal opposite attractive forces.

04. Build the solution (practice)
24 N; 3 N.

05. Use the idea (reason)
Equal forces; smaller moon usually has greater acceleration from that interaction.

06. Choose a method (mixed) *Review: Unit 1, Day 14; 12 course days later.*
Ground pushes the runner’s foot forward with 60 N. This external force can accelerate the runner; the partner forces act on different objects.

07. Exit check (exit)
2 N.

08. STRETCH: Apply it another way (stretch)
12×2/4=6 N; increased mass partly offsets the greater separation.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 1; 3 course days later.*
Direction changes but speed remains 5 m/s. Instantaneous perpendicular force does no work; velocity is not constant.

02. Explain a worked example (worked)
Center separation r then 2 r; each body's force drops from 20 to 5 N, with equal opposing forces within each pair.

03. Represent and interpret (model)
Distance is between centers; equal opposite attractive forces.

04. Solve and justify (practice)
$3\times 2/3^2=2/3$.

05. Analyze the reasoning (reason)
Equal forces; smaller moon usually has greater acceleration from that interaction.

06. Choose a method (mixed) *Review: Unit 1, Day 14; 12 course days later.*
Equal opposite 24 N forces. Craft acceleration 6 m/s²; asteroid 0.06 m/s² opposite. Compares this interaction’s contribution, or assumes other forces negligible.

07. Exit check (exit)
3 r for unchanged masses. If both triple, need 9 r to reach $F_0/9$.

08. STRETCH: Apply it another way (stretch)
Factor 9 in numerator: separation 3 r preserves force; $3\sqrt{2}$ r gives half force.